

yt-dlp: Download Youtube Videos

This article is part of in the series

yt-dlp/yt-dlp

A feature-rich command-line audio/video downloader



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yt-dlp is a feature-rich fork of youtube-dl that allows downloading videos from YouTube and many other video platforms. While its predecessor was groundbreaking, yt-dlp has evolved to offer superior download speeds, better platform support, and more robust features for managing video downloads. Whether you're an archivist, content creator, or just someone who wants to save videos for offline viewing, yt-dlp provides the flexibility and functionality you need. This guide covers installation, basic usage, advanced features, and best practices.

Help



Why Choose yt-dlp?

Unlike basic video downloaders, yt-dlp offers several advantages that make it stand out:

- Significantly faster download speeds through advanced techniques like segmented downloading
- Active maintenance and regular updates to handle platform changes
- Extensive format selection options for both video and audio
- Built-in support for hundreds of video platforms beyond just YouTube
- Robust error handling and retry mechanisms

Getting Started with Installation

Installing yt-dlp is straightforward across all major operating systems. The tool is available through various package managers, making it accessible to users regardless of their technical expertise. For Python users, pip installation provides the most up-to-date version, while system package managers offer stable releases with automatic updates.

Installation Methods Explained

Using pip (Python Package Manager)

```
pip install yt-dlp
```

When installing via pip (`pip install yt-dlp`), you'll get the latest version directly from the Python Package Index. This method works across all platforms and is particularly useful for users who want to stay on the cutting edge of features and fixes.

Using Package Managers

On macOS, (Homebrew):

```
brew install yt-dlp
```

On Windows (Chocolatey):

```
choco install yt-dlp
```

For macOS users, Homebrew (`brew install yt-dlp`) integrates yt-dlp into your system's package management, making updates and maintenance easier. Windows users benefit from Chocolatey's (`choco install yt-dlp`) automated installation process, which handles dependencies and path configuration automatically.

Using Package Managers

macOS users, (Homebrew):

```
brew install yt-dlp
```

Windows users (Chocolatey):

```
choco install yt-dlp
```

Understanding Video Quality and Formats

One of yt-dlp's most powerful features is its sophisticated format selection system. Modern video platforms often serve content in separate video and audio streams, which yt-dlp can automatically combine for optimal quality. When you request the "best" quality, yt-dlp intelligently selects the highest quality video and audio streams available, merging them seamlessly.

Downloading a Single Video

The simplest way to download a video is to provide the URL:

```
yt-dlp https://www.youtube.com/watch?v=example
```

Selecting Video Quality

Download the best quality video and audio:

```
yt-dlp -f "bestvideo+bestaudio" https://www.youtube.com/watch?v=example
```

Download specific quality (e.g., 1080p):

```
yt-dlp -f "bestvideo[height<=1080]+bestaudio" https://www.youtube.com/watch?v=example
```

Audio-Only Downloads

Extract audio in best quality:

```
yt-dlp -x --audio-format mp3 https://www.youtube.com/watch?v=example
```

Advanced Features

Downloading Playlists

Download all videos from a playlist:

```
yt-dlp https://www.youtube.com/playlist?list=example
```

Downloading specific videos from a playlist:

```
yt-dlp --playlist-start 1 --playlist-end 5 https://www.youtube.com/playlist?list=example
```

Custom Output Templates

The output template system in yt-dlp is remarkably flexible, allowing you to organize downloads based on any metadata the video provides. You can create sophisticated directory structures that automatically sort videos by creator, date, quality, or any combination of factors.

To specify a custom filename format:

```
yt-dlp -o "%(title)s-%(id)s.%(ext)s" https://www.youtube.com/watch?v=example
```

Organize downloads by channel:

```
yt-dlp -o "%(uploader)s/%(title)s.%(ext)s" https://www.youtube.com/watch?v=example
```

Download with Subtitles

For educational content or foreign language videos, yt-dlp's subtitle download capabilities are invaluable. The tool can download subtitles in multiple languages, including automatically generated captions, making content more accessible and useful for different audiences.

Downloading a video with all available subtitles:

```
yt-dlp --write-sub https://www.youtube.com/watch?v=example
```

Download specific language subtitles:

```
yt-dlp --write-sub --sub-lang en https://www.youtube.com/watch?v=example
```

Format Selection

List Available Formats

Show all available formats for a video:

```
yt-dlp -F https://www.youtube.com/watch?v=example
```

Specific Format Selection

Download a specific format using format code:

```
yt-dlp -f 137+140 https://www.youtube.com/watch?v=example
```

Format Selection Examples

Best video with best audio:

```
yt-dlp -f "bestvideo+bestaudio" https://www.youtube.com/watch?v=example
```

Best video under 1080p with best audio:

```
yt-dlp -f "bestvideo[height<=1080]+bestaudio" https://www.youtube.com/watch?v=example
```

Advanced Configuration

Using Configuration Files

Create a configuration file (`~/.config/yt-dlp/config` or `%APPDATA%/yt-dlp/config`):

```
# Default format
-f "bestvideo+bestaudio"
# Always extract audio
-x
--audio-format mp3
# Output template
-o "%(title)s.%(ext)s"
```

Rate Limiting and Throttling

Download with rate limit:

```
yt-dlp --limit-rate 1M https://www.youtube.com/watch?v=example
```

Batch Downloads

Download from a file containing URLs:

```
yt-dlp -a urls.txt
```

Authentication

Using Cookies

Download with browser cookies:

```
yt-dlp --cookies-from-browser firefox https://www.youtube.com/watch?v=example
```

Using Credentials

Use username and password:

```
yt-dlp -u username -p password https://www.youtube.com/watch?v=example
```

Updating and Maintenance

Regular updates are crucial as video platforms frequently change their systems. yt-dlp's update command (`yt-dlp -U`) should be run periodically to ensure you have the latest features and platform support.

Update yt-dlp

Update to the latest version:

```
yt-dlp -U
```

Debug Information

Show debug information:

```
yt-dlp --verbose https://www.youtube.com/watch?v=example
```

Best Practices and Workflow Optimization

Creating an Efficient Download Strategy

When working with yt-dlp, it's important to develop a consistent strategy for organizing downloads. Consider creating a standardized directory structure and naming convention that works for your needs. For example, organizing by content type, creator, and date can make managing large collections much easier.

Resource Management

While yt-dlp is efficient, downloading high-quality videos can be resource-intensive. Consider using rate limiting on slower connections or when downloading multiple videos simultaneously. This helps prevent network saturation and ensures stable downloads.

Automation and Integration

yt-dlp can be easily integrated into larger workflows through scripts and automation tools. Consider setting up automated downloads for regularly updated content, or integrate yt-dlp into your content management system for streamlined media handling.

1. Using Output Templates

```
# Organize by type and quality
```

```
yt-dlp -o "%(uploader)s/%(height)sp/%(title)s.%(ext)s" URL
```

The provided example
`yt-dlp -o "%(uploader)s/%(height)sp/%(title)s.%(ext)s"`
 demonstrates a well-structured output template. This template creates a directory structure based on the uploader and video resolution (height). Organizing like this allows you to easily browse and filter your downloaded videos.

2. Creating Aliases

```
# For bash/zsh
```

```
alias
```

```
yt="yt-dlp -f 'bestvideo+bestaudio' -o '%(title)s.%(ext)s'"
```

This example alias downloads the best available video and audio tracks and saves them as `%(title)s.%(ext)s` in the current directory.

3. **Using Configuration Files** Create a config file with your most used options to avoid typing them repeatedly.

Common Issues and Solutions

Network issues, region restrictions, and platform changes can sometimes interrupt downloads. yt-dlp includes robust retry mechanisms and can often work around temporary issues automatically. Understanding these features helps ensure successful downloads even under challenging conditions.

Network Issues

If you encounter network errors, use this code to retry:

```
yt-dlp --fragment-retries infinite --retries infinite URL
```

Format Selection Issues

If the default format selection fails you can use this code:

```
yt-dlp -f "bestvideo[ext=mp4]+bestaudio[ext=m4a]/best[ext=mp4]/best" URL
```


Tips and Tricks

1. Download Thumbnails

```
yt-dlp --write-thumbnail URL
```

2. Extract Metadata

```
yt-dlp --write-info-json URL
```

3. Download with Description

```
yt-dlp --write-description URL
```

In summation, yt-dlp is a powerful tool for downloading video content. By understanding its features and options, you can efficiently download and manage video content from various platforms.

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